

Compensating Wage Differentials

ECON 300 - Labour Economics

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Learning Objectives

- Explain how wages reflect job attributes such as **working conditions, safety risks, shift work, pensions, and health benefits**.
- Understand the meaning of **compensating wage differentials**: wage differences that compensate workers for undesirable job characteristics.
- Identify when market-determined workplace risk may be **efficient** and when market outcomes may fail.
- Evaluate how **safety regulation** affects workers, firms, and total welfare.
- Understand the principle of “**no free lunch**”: desirable job amenities are often accompanied by **lower wages**.
- Use wage premiums for risky jobs to infer the **value of a statistical life (VSL)**.

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Big Idea of the Chapter

Core Insight

Wages do not only pay for **skills and productivity**. They also reflect **job characteristics** such as risk, discomfort, flexibility, and fringe benefits.

- Workers trade off **money wages** against **non-wage job characteristics**.
- Firms choose combinations of wages and job attributes that minimize cost or maximize profit.
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Purposes of Wages and Wage Structures

Definition: Wage Structure

The **wage structure** is the hierarchy or grid of wages across jobs. It includes wage **premiums** linked to worker skills, job tasks, working conditions, and amenities.

- **Wage–productivity link:** In competitive settings, wages must reflect marginal productivity.
- **Compensation role:** Jobs with unpleasant or risky conditions often require higher wages.
- **Organizational role:** Wages also help with motivation, fairness, recruitment, and retention.
- **Example:** Miners or offshore workers often earn more than office workers partly because of risk and harsh conditions.

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Why Can Similar Workers Earn Different Wages?

- Even equally skilled workers may earn different wages if jobs differ in:
 - workplace safety,
 - scheduling convenience,
 - pensions and health insurance,
 - location and commuting burden,
 - employment stability.
- These differences are called **compensating wage differentials**.
- Thus, wage inequality may reflect both **worker characteristics** and **job characteristics**.

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Smith's Compensating Differentials: What Gets Priced?

Historical Insight

Adam Smith argued that wage differences across occupations reflect the **advantages and disadvantages** attached to different jobs.

- Agreeableness or disagreeableness of tasks
- Training cost and difficulty
- Job stability and probability of success
- Responsibility, trust, and authority
- Safety and risk

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- Pleasant or flexible jobs may pay **less** because workers value the amenity.
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Isoprofit Schedules (Firms' Side)

Definition: Isoprofit Curve

An **isoprofit schedule** shows all combinations of **wage** and **safety** that yield the same profit for the firm.

- Firms can compensate workers by paying higher wages or by providing safer conditions.
- Improving safety may require spending on equipment, training, or monitoring.
- Lower isoprofit curves correspond to **higher profits**.
- The slope reflects the firm's trade-off between safety spending and wage costs.

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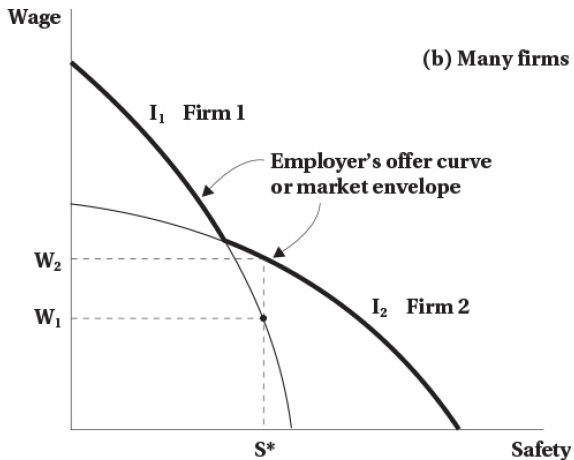
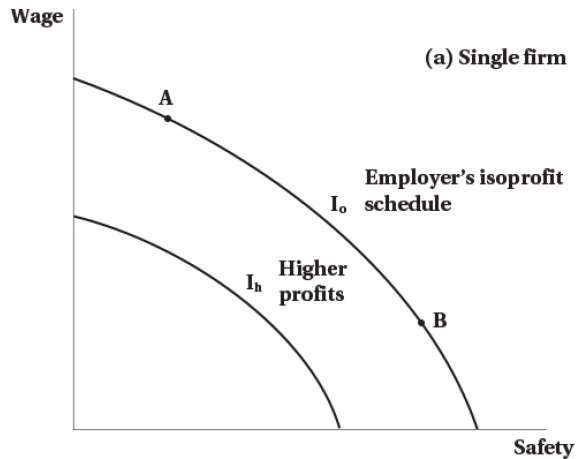
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Figure 8.1: Employer's Isoprofit Schedules and Offer Curve



Explaining Figure 8.1

- Each isoprofit curve shows wage–safety combinations that give the same profit.
- Curves farther from the origin generally imply **lower profits**.
- Tangency between isoprofit schedules and workers' preferences determines feasible offers.
- The **offer curve** traces the combinations of wage and safety that firms are willing to offer as worker preferences differ.

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Different Safety Technologies Across Firms

- Firms differ in the **cost of providing safety**.
- Some firms can improve safety cheaply; others face high safety costs.
- Therefore, even for the same profit level, firms may have **different isoprofit curves**.
- Example: A modern factory may provide safety more cheaply than an old mine.

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Definition: Indifference Curve

An **indifference curve** shows combinations of **wage** and **safety** that give the worker the same utility.

- Workers prefer:
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- More risk-averse workers require larger wage premiums to accept danger.
- Thus, worker heterogeneity is central to compensating wage theory.

Worker Preferences and Indifference Curves

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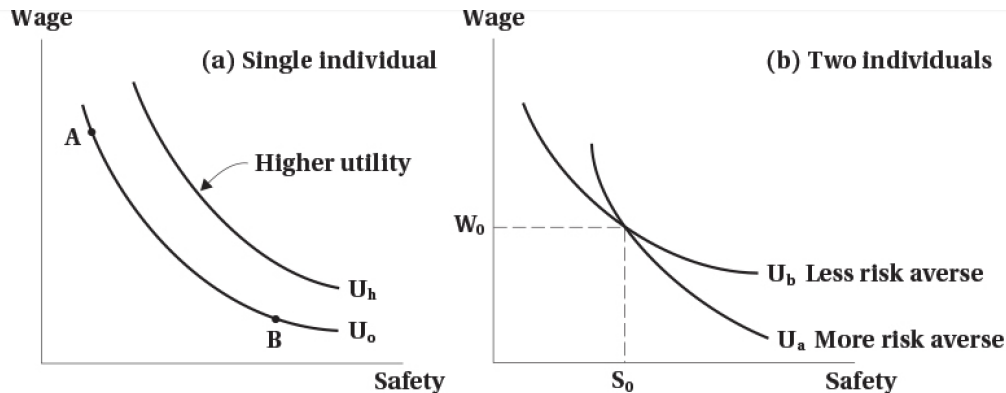
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Figure 8.2: Worker Indifference Curves



Explaining Figure 8.2

- Each indifference curve shows wage–safety bundles that leave the worker equally well off.
- Higher indifference curves imply **higher utility**.
- A steep indifference curve means the worker values safety highly and requires a large wage increase to accept more risk.
- A flatter indifference curve means the worker is more willing to trade safety for wages.

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Equilibrium with One Firm and One Worker

- The optimal wage–safety bundle occurs at the **tangency** of:
 - the firm's isoprofit curve, and
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- At tangency:
 - the worker cannot be made better off without lowering firm profit,
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Many Firms, Many Workers (Sorting)

Definition: Sorting

Sorting occurs when different workers choose different jobs based on their preferences, and firms choose different workplace attributes based on their technologies.

- Risk-tolerant workers sort into dangerous jobs with higher wages.
- Highly risk-averse workers sort into safer jobs with lower wages.
- Firms with low-cost safety technology offer safer jobs.
- This produces a market **wage–safety locus**.

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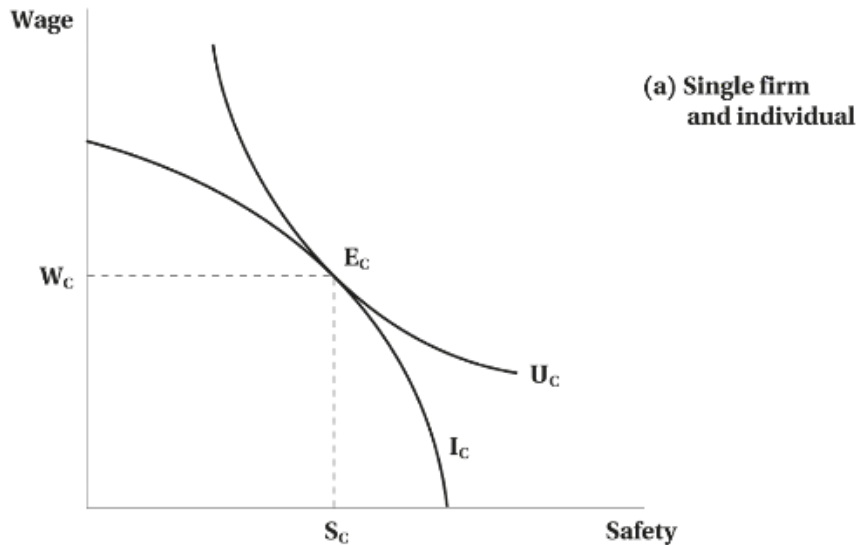
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Figure 8.3: Market Equilibrium



Characteristics of the Wage–Safety Locus

- **Negative slope:** Lower safety must be compensated by higher wages.
- The shape depends on:
 - worker preferences toward risk,
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- The locus summarizes equilibrium outcomes across many firms and many workers.

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Regulation in Competitive Markets

- In a frictionless competitive market, workers choose their preferred wage–safety trade-off.
- If the market already reflects informed choices, a safety mandate may:
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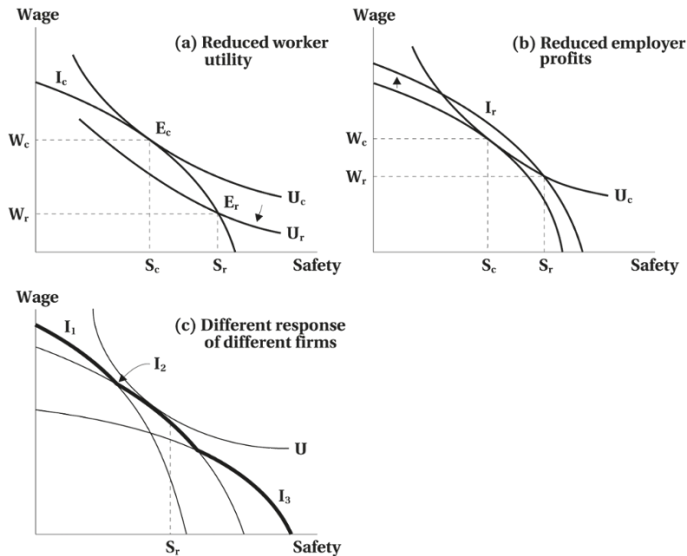
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Figure 8.4: Responses to a Safety Standard



Explaining Safety Standards

- A mandated safety standard pushes firms toward safer jobs.
- This usually reduces the need for compensating wage premiums.
- Wages may fall because workers are receiving more safety instead of money.
- Whether regulation increases welfare depends on whether the original market outcome was efficient.

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Worked Example: Attracting Doctors to Rural Areas

Two doctors can earn \$100,000 in a city.

- Dr. Brown values city amenities at \$90,000:

$$100,000 + 90,000 = 190,000$$

so she needs about \$190,000 to be indifferent to moving rural.

- Dr. Lewis values city amenities at only \$40,000:

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so she would move for a lower rural salary.

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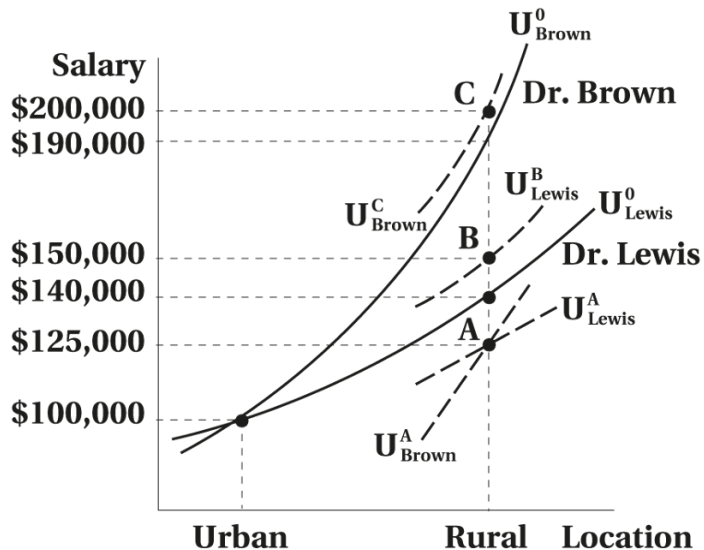
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Location Decision Illustration



Example: Other Compensating Differentials

- **Night shifts:** workers often receive a shift premium.
- **Remote locations:** jobs in northern or isolated areas often pay more.
- **Flexible work hours:** workers may accept lower wages in exchange for flexibility.
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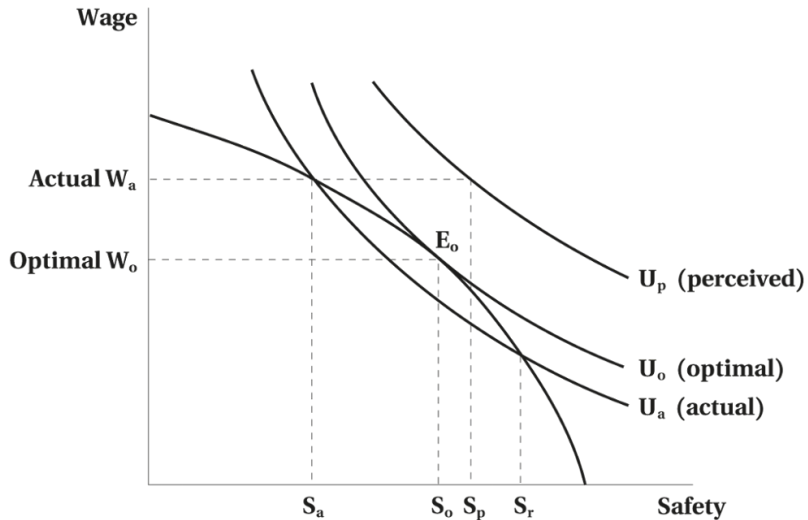
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Figure 8.5: Effect of Imperfect Information



Why Regulate?

- **Information failures:** workers may not know the true risks.
- **Externalities:** workplace injuries may impose costs on families, insurers, or taxpayers.
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Wage–Risk Trade-offs and Value of Life

- Risk premia tend to rise with the severity of workplace hazards.
- Workers' compensation or insurance can reduce required wage premia.
- Economists estimate the **wage–risk gradient** and use it to infer VSL.
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Desirable vs. Undesirable Job Characteristics

- Empirical evidence often finds wage trade-offs for:
 - health insurance,
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Family-Friendly Work Practices

- Family-friendly policies often behave like job amenities.
- Workers may accept lower wages for:
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- Evidence is mixed for some policies, especially childcare support and work-from-home arrangements.

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- Policies should account for:
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